

Name: _____

Class _____



HSC Assessment Task 1 2023-2024

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 65 minutes
- Board approved calculators may be used.
- Write using black or blue pen
- All necessary working should be shown in all questions
- Write your student number at the top of every page of your answer sheets.
- NESAs Reference Sheet is provided

Total marks – 40

Attempt all questions

Section A – Answer on the Multiple Choice Answer Sheet

Section B - Start each question on a new sheet of paper.

NESA Reference Sheet is provided

This paper MUST NOT be removed from the examination room

Section A Multiple Choice (5 Marks)

QUESTION 1

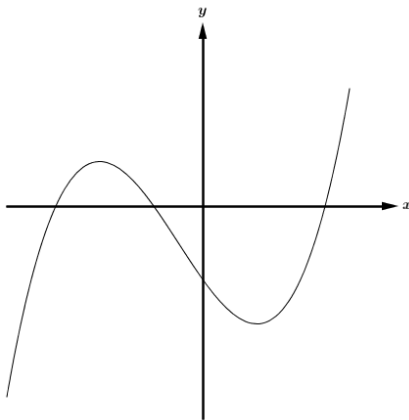
Which of the following step is not necessary to prove a statement to be true for all positive integer values n by mathematical induction?

- A. Prove that the statement is true for $n = 1$.
- B. Prove that if the statement is true for some case $n = k$, then the statement is also true for $n = k + 1$.
- C. Check that the statement is true for a very large value of n .
- D. None of the above.

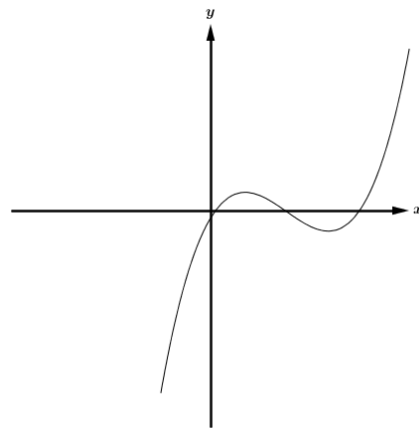
QUESTION 2

Consider the polynomial $P(x) = ax^3 + bx^2 + cx - 6$, where $a, b > 0$. Which of the following is a potential graph of $y = P(x)$?

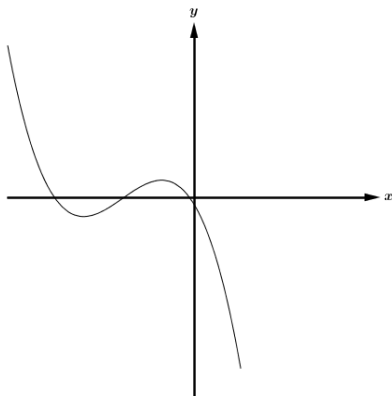
A.



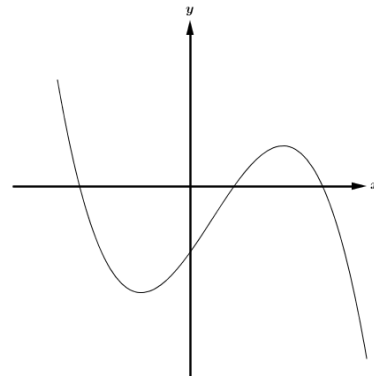
B.



C.



D.



QUESTION 3

Which of the following will have a non-zero coefficient in the expansion of $\left(a^2 + \frac{1}{a}\right)^{23}$?

- A. a^{-9}
- B. a^{-8}
- C. a^{-7}
- D. a^{-6}

QUESTION 4

Consider the expansion of $\left(2x^3 + \frac{1}{x^2}\right)^n$, $n \in \mathbb{Z}^+$. If it has a non-zero constant term, what is the minimum possible value of n ?

- A. 3
- B. 5
- C. 8
- D. 10

QUESTION 5

Given $\sin \beta - \cos \beta = \frac{4}{3}$, what is the value of $\sin 2\beta$?

- A. $-\frac{2}{9}$
- B. $-\frac{7}{9}$
- C. $\frac{2}{9}$
- D. $\frac{7}{9}$

SECTION B

QUESTION 6 Start a new page (5 marks)

Marks

a) Let α, β and γ be the roots of $2x^3 - 4x + 7$.

i. Find the values of $\alpha\beta + \alpha\gamma + \beta\gamma$ and $\alpha + \beta + \gamma$. 2

ii. Hence or otherwise, find the value of $\alpha^3 + \beta^3 + \gamma^3$. 2

b) How many terms are there in the expansion of $(3a + 2b)^{90209}$? 1

QUESTION 7 Start a new page (5 marks)

Marks

a) Prove by mathematical induction that for $n \in \mathbb{Z}^+$ 3

$$a + ar + ar^2 + \dots + ar^n = a \left(\frac{r^{n+1} - 1}{r - 1} \right)$$

if $r \neq 1$.

b) Given $(1 + x + x^2)^6 = a_0 + a_1x + a_2x^2 + \dots + a_{12}x^{12}$, find the value of: 2

$$a_2 + a_4 + a_6 + a_8 + a_{10} + a_{12}.$$

QUESTION 8 Start a new page (6 marks)

Marks

a) Suppose that b and d are real numbers and $d \neq 0$. Consider the polynomial $P(x) = x^4 + bx^2 + d$, $P(x) = 0$ has a double root at $x = \alpha$.

i. Find the remainder when $P(x)$ is divided by $(x - 1)$. 1

ii. Show that $\alpha^2 = -\frac{b}{2}$. 2

iii. Hence or otherwise, show that $\frac{b^2}{4} = d$. 2

QUESTION 9 Start a new page (5 marks)**Marks**

- a) Prove by mathematical induction that $23n^3 + 97n$ is divisible by 6 for all integers $n \geq 1$. **4**
- b) Fully simplify $(2x + 1)^5 - 5(2x + 1)^4 + 10(2x + 1)^3 - 10(2x + 1)^2 + 5(2x + 1) - 1$. **1**

QUESTION 10 Start a new page (5 marks)**Marks**

- a) Use the t -formulae to solve the equation $2 \sin \frac{x}{2} + \cos \frac{x}{2} = 1$ for $x \in [0, 8\pi]$. **3**
- b) Prove that $\sin\left(\theta + \frac{\pi}{6}\right) \sin\left(\theta - \frac{\pi}{6}\right) = \sin^2 \theta - \frac{1}{4}$. **2**

QUESTION 11 Start a new page (5 marks)**Marks**

- a)
- i. Express $\sin \theta + \sin\left(\theta + \frac{\pi}{3}\right)$ in the form $R \sin(\theta + \alpha)$, where $R > 0$, and $0 < \alpha < \frac{\pi}{2}$. Show all necessary working. **3**
- ii. Hence, or otherwise sketch the graph $y = \sin \theta + \sin\left(\theta + \frac{\pi}{3}\right)$, for $0 \leq \theta \leq 2\pi$, showing important features. **2**

QUESTION 12 Start a new page (5 marks)**Marks**

- a) Use the binomial expansion to prove that:
- $$n \times 3^{n-1} = \binom{n}{1}(1) + \binom{n}{2}(2 \times 2) + \binom{n}{3}(3 \times 2^2) + \cdots \binom{n}{n}(n \times 2^{n-1}).$$
- 2**
- b) Given $\sin \alpha$, $\sin \beta$ and $\sin \gamma$ are the solutions to the equation to $8x^3 - 6x - \sqrt{5} = 0$, find the value of $\cot^2 \alpha + \cot^2 \beta + \cot^2 \gamma$. **3**

End of examination

Multiple Choice:

1. (C)
2. Degree 3, positive leading coefficient therefore Option (A) or (B)

$$\begin{aligned}\text{Sum of roots} &= -\frac{b}{a} \\ &< 0\end{aligned}$$

Sum of all x – intercepts should be negative value.

Thus, rules out Option (B)

Therefore answer is (A)

3. Start from a^{-23} , the powers of a goes up by 3, so $-23, -20, -17, -14, -11, -8, -5, -2, 1, 4, 7, 10, 13, 16, 19, 22, 25$
 $-7 \therefore$ (B)
4. Constant term $\Rightarrow (x^3)^r(x^{-2})^{n-r} = x^{5r-2n} = x^0$
 $\therefore 5r - 2n = 0$
 $n = 5$ and $r = 2$ is the first possibility
(B)

5. (B)

MATHEMATICS EXT 1: Question 6

Suggested Solutions	Marks	Marker's Comments
a i. Let $p(x) = 2x^3 + 0x^2 - 4x + 7$ $\sum x = -\frac{b}{a}$ $\therefore \alpha + \beta + \gamma = -\frac{0}{2} = 0$ _____ $\sum \alpha\beta = \frac{c}{a}$ $\alpha\beta + \alpha\gamma + \beta\gamma = \frac{-4}{2} = -2$ _____	① ①	
ii. Since α, β, γ are roots of $p(x)$, then $p(\alpha) = 0 \Rightarrow 2\alpha^3 - 4\alpha + 7 = 0$ ① $p(\beta) = 0 \Rightarrow 2\beta^3 - 4\beta + 7 = 0$ ② $p(\gamma) = 0 \Rightarrow 2\gamma^3 - 4\gamma + 7 = 0$ ③ ① + ② + ③ : $\therefore 2(\alpha^3 + \beta^3 + \gamma^3) - 4(\alpha + \beta + \gamma) + 21 = 0$ $2(\alpha^3 + \beta^3 + \gamma^3) = -21$ $\alpha^3 + \beta^3 + \gamma^3 = -\frac{21}{2}$ _____	①	
b. 90210 terms _____	①	

a) To prove by MI $a + ar + ar^2 + \dots + ar^n = \frac{a(r^{n+1} - 1)}{r - 1}$

Solution:

$$n=1, \text{ LHS} = a + ar^1 = a(r+1)$$

$$\text{RHS} = \frac{a(r^{1+1} - 1)}{r - 1} = \frac{a(r^2 - 1)}{r - 1} = \frac{a(r+1)(r-1)}{r-1} = a(r+1) \quad \left. \vphantom{\frac{a(r^2 - 1)}{r - 1}} \right\} \text{Im}$$

$\therefore$ true

Assume true for $n=k$, $k \in \mathbb{Z}^+$

$$a + ar + ar^2 + \dots + ar^k = \frac{a(r^{k+1} - 1)}{r - 1}$$

RTP for $n=k+1$

$$a + ar + ar^2 + \dots + ar^k + ar^{k+1} = \frac{a(r^{k+2} - 1)}{r - 1}$$

$$\text{LHS} = \frac{a(r^{k+1} - 1)}{r - 1} + ar^{k+1} \quad (\text{By assumption}) \quad \left. \vphantom{\frac{a(r^{k+1} - 1)}{r - 1}} \right\} \text{Im}$$

$$= \frac{a(r^{k+1} - 1) + (r - 1)ar^{k+1}}{r - 1}$$

$$= \frac{a \cancel{r^{k+1}} - a + ar^{k+2} - a \cancel{r^{k+1}}}{r - 1} \quad \left. \vphantom{\frac{a \cancel{r^{k+1}} - a + ar^{k+2} - a \cancel{r^{k+1}}}{r - 1}} \right\} \text{Im}$$

$$= \frac{a(r^{k+2} - 1)}{r - 1} = \text{RHS}$$

Conclusion: By the principle of Maths Induction, therefore true for all $n \in \mathbb{Z}^+$

b) $(1+x+x^2)^6 = a_0 + a_1x + a_2x^2 + \dots + a_{12}x^{12}$ find the value of $a_2 + a_4 + a_6 + a_8 + a_{10} + a_{12}$

Solution when $x=0$, $1^6 = a_0 \Rightarrow a_0 = 1$

$$\text{when } x=-1 \quad (1-x+x^2)^6 = a_0 - a_1 + a_2 - a_3 + \dots + a_{12} \quad \left. \vphantom{(1-x+x^2)^6} \right\} \text{Im}$$

$$x = x - (a_1 + a_3 + a_5 + a_7 + a_9 + a_{11}) + (a_2 + a_4 + \dots + a_{12})$$

$$\therefore a_1 + a_3 + a_5 + \dots + a_{11} = a_2 + a_4 + \dots + a_{12}$$

$$\text{when } x=1 \quad (1+1+1)^6 = a_0 + a_1 + a_2 + \dots + a_{11} + a_{12}$$

$$3^6 = 1 + 2(a_2 + a_4 + \dots + a_{12}) \quad \left. \vphantom{3^6} \right\} \text{Im}$$

$$729 - 1 = 2(a_2 + a_4 + \dots + a_{12})$$

$$\underline{\underline{364}} = a_2 + a_4 + \dots + a_{12}$$

Method 2 Need to see the development of the double summation

$$(1+x+x^2)^6 = [1+x(1+x)]^6 = \sum_{r=0}^6 \binom{6}{r} x^{6-r} (1+x)^{6-r}$$

$$= \binom{6}{0} x^6 (1+x)^6 + \binom{6}{1} x^5 (1+x)^5 + \binom{6}{2} x^4 (1+x)^4 + \binom{6}{3} x^3 (1+x)^3$$

$$+ \binom{6}{4} x^2 (1+x)^2 + \binom{6}{5} x (1+x) + \binom{6}{6}$$

Coeff of positive even powers of $x \Rightarrow x^2, x^4, x^6, x^8, x^{10}, x^{12}$

$$\binom{6}{0} [\binom{6}{0} + \binom{6}{2} + \binom{6}{4} + \binom{6}{6}] + \binom{6}{1} [\binom{5}{1} + \binom{5}{3} + \binom{5}{5}] + \binom{6}{2} [\binom{4}{0} + \binom{4}{2} + \binom{4}{4}]$$

$$+ \binom{6}{3} [\binom{3}{1} + \binom{3}{3}] + \binom{6}{4} [\binom{2}{0} + \binom{2}{2}] + \binom{6}{5}$$

1 m

$$= 1(1+15+15+1) + 6(5+10+1) + 15(1+6+1) + 20(3+1) + 15(1+1) + 6$$

$$= 22 + 96 + 120 + 80 + 10 + 30 + 6 = \underline{\underline{364}}$$

1 m

Method 3 Need to see the double summation

$$(1+x+x^2)^6 = [(1+x)+x^2]^6 = \sum_{r=0}^6 \binom{6}{r} (1+x)^{6-r} (x^2)^r$$

$$= \sum_{r=0}^6 \binom{6}{r} \sum_{k=0}^{6-r} \binom{6-r}{k} x^k x^{2r} = \sum_{r=0}^6 \binom{6}{r} \sum_{k=0}^{6-r} \binom{6-r}{k} x^{2r+k}$$

General term is $\binom{6}{r} \binom{6-r}{k} x^{2r+k}$ $0 \leq r \leq 6, 0 \leq k+r \leq 6$

$2r+k=2$

k	r
0	1
2	0

$$\binom{6}{1} \binom{5}{0} + \binom{6}{0} \binom{6}{2}$$

$$= 21$$

$2r+k=4$

k	r
0	2
2	1
4	0

$$\binom{6}{2} \binom{4}{0} + \binom{6}{1} \binom{5}{2} + \binom{6}{0} \binom{6}{4}$$

$$= 90$$

$2r+k=6$

k	r
0	3
2	2
4	1
6	0

$$\binom{6}{3} \binom{3}{0} + \binom{6}{2} \binom{4}{2} + \binom{6}{1} \binom{5}{4} + \binom{6}{0} \binom{6}{6}$$

$$= 141$$

$2r+k=8$

k	r
0	4
2	3
4	2

$$\binom{6}{4} \binom{2}{0} + \binom{6}{3} \binom{3}{2} + \binom{6}{2} \binom{4}{4}$$

$$= 90$$

$2r+k=10$

k	r
0	5
2	4

$$\binom{6}{5} \binom{1}{0} + \binom{6}{4} \binom{2}{2}$$

$$= 21$$

k	r
0	6

$$\binom{6}{6} \binom{0}{0}$$

$$= 1$$

Total $2(21+90) + 141 + 1$

$$= \underline{\underline{364}}$$



Student Number _____

QUESTION 8

$$\begin{aligned} \text{i) } P(1) &= 1^4 + b \cdot 1^2 + d \\ &= 1 + b + d \end{aligned}$$

$1 + b + d$ is the remainder upon division by $(x-1)$ ✓

$$\text{ii) } P(x) = 0 \Rightarrow x^4 + bx^2 + d = 0 \quad (1)$$

$$P'(x) = 0 \Rightarrow 4x^3 + 2bx = 0 \quad (2)$$

$$\text{from (2): } 4x^3 + 2bx = 0$$

$$2x(2x^2 + b) = 0$$

$$x = 0 \text{ or } 2x^2 = -b$$

$$x^2 = -\frac{b}{2}$$

$$\begin{aligned} \text{if } x=0 \quad 0^4 + b \cdot 0^2 + d &= 0 \quad \text{from (1)} \\ d &= 0 \end{aligned}$$

$$\text{but } d \neq 0 \quad \therefore x^2 = -\frac{b}{2} \text{ only}$$

✓ solving $P'(x) = 0$
↓
(applying theory of multiple roots)

✓ rejecting
 $x=0$

iii) Substituting this result into (1):

$$\left(-\frac{b}{2}\right)^2 + b\left(-\frac{b}{2}\right) + d = 0 \quad \text{✓ substitute into}$$

$$\frac{b^2}{4} - \frac{b^2}{2} + d = 0 \quad \text{P(x)}$$

$$\frac{b^2}{4} = d \quad \downarrow \text{ simplify.}$$

If relationships between roots were attempted then no mark was given

for just writing out $\alpha + \alpha + \beta + \gamma = \dots$

$$\alpha^2 + 2\alpha\beta + 2\alpha\gamma + \beta\gamma = \dots$$

etc

First mark was for an attempt to solve simultaneously via substitution or elimination.

Question 9

a) Test for $n = 1$

$$\begin{aligned} 23(1)^3 + 97(1) &= 120 \\ &= 6 \times 20 \leftarrow (\text{MUST SHOW!!!!}) \end{aligned}$$

$\therefore$ the statement is true for $n = 1$ (1 mark)

Assume the statement is true for $n = k, k \in \mathbb{Z}^+$

$\therefore 23k^3 + 97k = 6M_1$ where $M_1 \in \mathbb{Z}$ (1 mark for interpreting what the assumption means)

Test for $n = k + 1$

RTP: $23(k + 1)^3 + 97k = 6M_2$ where $M_2 \in \mathbb{Z}^+$

$$\begin{aligned} \text{LHS} &= 23(k + 1)^3 + 97(k + 1) \\ &= 23(k^3 + 3k^2 + 3k + 1) + 97(k + 1) \\ &= (23k^3 + 97k) + (69k^2 + 69k + 23 + 97) \\ &= 6M_1 + (69k^2 + 69k + 120) \quad (\text{By Assumption}) \end{aligned}$$

(1 mark for using the assumption correctly, you do **NOT** get the mark if you substituted and ended up with a fraction)

$$\begin{aligned} &= 6M_1 + 120 + 69k(k + 1) \\ &= 6M_1 + 120 + 69 \times (2P) \quad \text{where } P \in \mathbb{Z} \end{aligned}$$

(the product of 2 consecutive integers must be even)

$$\begin{aligned} &= 6M_1 + 120 + 138P \\ &= 6(M_1 + 20 + 23P) \\ &= 6M_2 \quad (\text{Since } M_1 + 20 + 23P \in \mathbb{Z}) \end{aligned}$$

$\therefore$ if the statement is true for $n = k$, then it is true for $n = k + 1$

$\therefore$ the statement is true for $n = \mathbb{Z}^+$ by the principle of mathematical induction. (1 mark)

(Note: There is 1 mark in this question allocated to all the small details that is required in a mathematical induction proof, such as defining the number set of any variables introduced, making clear that 120 is a multiple of 6, writing by assumption when a substitution is made, using the write mathematical language like assume the statement is true for $n=k$. If you wrote words that convey an incorrect mathematical logic you will only receive 3 out of 4 as a maximum)

$$\begin{aligned} \text{b) } &(2x + 1)^5 - 5(2x + 1)^4 + 10(2x + 1)^3 - 10(2x + 1)^2 + 5(2x + 1) - 1 \\ &= (2x + 1 - 1)^5 \\ &= (2x)^5 \\ &= 32x^5 \quad (1 \text{ mark}) \end{aligned}$$

QUESTION 10 Start a new page (5 marks)

Marks

a) Use the t -formulae to solve the equation $2 \sin \frac{x}{2} + \cos \frac{x}{2} = 1$ for $x \in [0, 8\pi]$. 3

b) Prove that $\sin\left(\theta + \frac{\pi}{6}\right) \sin\left(\theta - \frac{\pi}{6}\right) = \sin^2 \theta - \frac{1}{4}$. 2

10.

a)

$$\begin{aligned} \text{Let } t &= \tan \frac{x}{4} & 0 \leq x \leq 8\pi \\ & & 0 \leq \frac{x}{4} \leq 2\pi \\ \therefore \frac{4t}{1+t^2} + \frac{1-t^2}{1+t^2} &= 1 \quad \checkmark \\ 4t + 1 - t^2 &= 1 + t^2 \quad \checkmark \\ 2t^2 - 4t &= 0 \\ 2t(t-2) &= 0 \\ t = 0 \text{ or } t &= 2 \\ \therefore \tan \frac{x}{4} &= 0 \text{ or } \tan \frac{x}{4} = 2 \quad \checkmark \\ \therefore \frac{x}{4} &= 0, \pi, 2\pi, 1.11 \text{ or } 4.25 \\ \therefore x &= 0, 4\pi, 8\pi, 4.44 \text{ or } 17.0 \quad \checkmark \end{aligned}$$

b)

$$\begin{aligned} \text{LHS} &= \left(\sin \theta \cos \frac{\pi}{6} + \cos \theta \sin \frac{\pi}{6} \right) \left(\sin \theta \cos \frac{\pi}{6} - \cos \theta \sin \frac{\pi}{6} \right) \\ &= \sin^2 \theta \cos^2 \frac{\pi}{6} - \cos^2 \theta \sin^2 \frac{\pi}{6} \quad \checkmark \\ &= \frac{3}{4} \sin^2 \theta - \frac{1}{4} \cos^2 \theta \\ &= \frac{3}{4} \sin^2 \theta - \frac{1}{4} (1 - \sin^2 \theta) \\ &= \frac{3}{4} \sin^2 \theta + \frac{1}{4} \sin^2 \theta - \frac{1}{4} \quad \checkmark \\ &= \sin^2 \theta - \frac{1}{4} \end{aligned}$$

$$\begin{aligned} \text{RHS} &= (\sin \theta)^2 - \frac{1}{4} = \left(\sin \theta + \frac{1}{2} \right) \left(\sin \theta - \frac{1}{2} \right) = \left(\sin \theta + \sin \frac{\pi}{6} \right) \left(\sin \theta - \sin \frac{\pi}{6} \right) = \\ &= 2 \sin \left(\frac{\theta + \frac{\pi}{6}}{2} \right) \cos \left(\frac{\theta - \frac{\pi}{6}}{2} \right) \times 2 \sin \left(\frac{\theta - \frac{\pi}{6}}{2} \right) \cos \left(\frac{\theta + \frac{\pi}{6}}{2} \right) = 2 \sin \left(\frac{\theta + \frac{\pi}{6}}{2} \right) \cos \left(\frac{\theta - \frac{\pi}{6}}{2} \right) \times 2 \sin \left(\frac{\theta - \frac{\pi}{6}}{2} \right) \cos \left(\frac{\theta + \frac{\pi}{6}}{2} \right) = \\ &= \sin \left(\theta + \frac{\pi}{6} \right) \sin \left(\theta - \frac{\pi}{6} \right) \quad \checkmark \end{aligned}$$

Suggested Solutions

Marks

Marker's Comments

11ai) $\sin \theta + \sin \left(\theta + \frac{\pi}{3} \right)$ in the form $R \sin(\theta + \alpha)$ $R > 0$, $0 < \alpha < \frac{\pi}{2}$

$$\sin \theta + \sin \left(\theta + \frac{\pi}{3} \right) = R \left[\sin(\theta + \alpha) \right]$$

working

LHS

$$\sin \theta + \sin \theta \cos \frac{\pi}{3} + \cos \theta \sin \frac{\pi}{3}$$

$$= \frac{3}{2} \sin \theta + \frac{\sqrt{3}}{2} \cos \theta$$

equate

RHS

$$R \left[\sin \theta \cos \alpha + \cos \theta \sin \alpha \right]$$

$$\left. \begin{aligned} R \cos \alpha &= \frac{3}{2} & \text{--- i} \\ R \sin \alpha &= \frac{\sqrt{3}}{2} & \text{--- ii} \end{aligned} \right\} \text{square both sides}$$

$$\left. \begin{aligned} R^2 \cos^2 \alpha &= \frac{9}{4} \\ R^2 \sin^2 \alpha &= \frac{3}{4} \end{aligned} \right\} \text{add}$$

$$R^2 (\cos^2 \alpha + \sin^2 \alpha) = 3$$

$$R = \sqrt{3} \quad (R > 0)$$

sub R into ii

$$\sin \alpha = \frac{\sqrt{3}}{\sqrt{3}(2)}$$

$$\alpha = \frac{\pi}{6} \quad 0 < \alpha < \frac{\pi}{2}$$

$$\therefore R \sin(\theta + \alpha) = \sqrt{3} \sin\left(\theta + \frac{\pi}{6}\right)$$

11a ii)

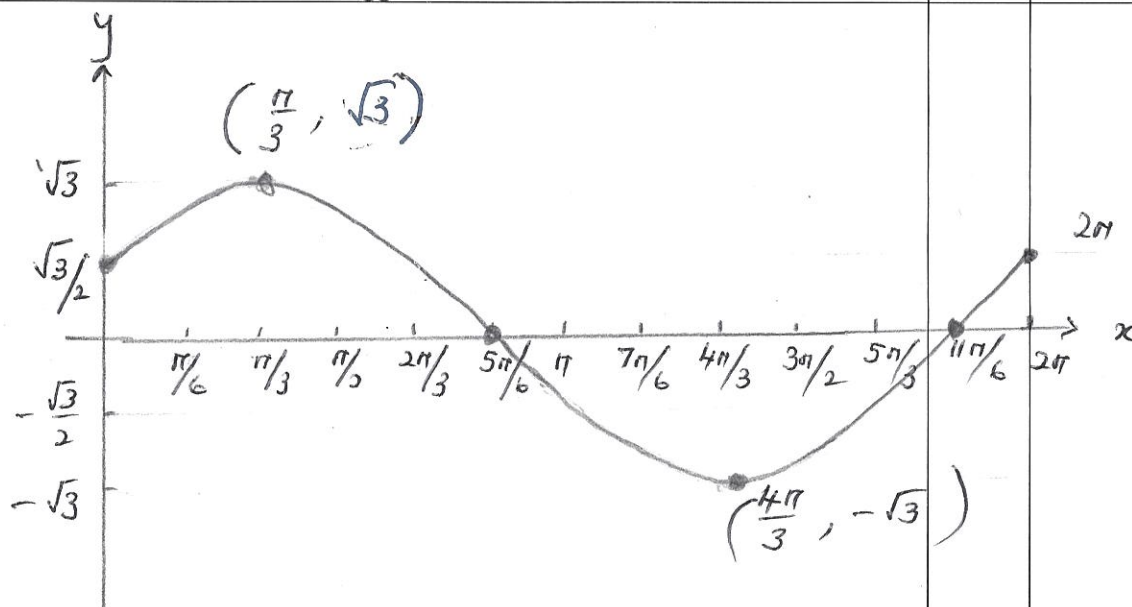
MATHEMATICS Extension 1. Question 11

Task 1 2024

Suggested Solutions

Marks

Marker's Comments



- * - min pt
- max pt
- x int
- end points
- scale - consistent for x & y
- shape
- domain
- end points not curved (not max t.p)

> 4 mistakes - 0

< 4 mistakes - 1

$$\sqrt{3} \sin\left(\theta + \frac{\pi}{6}\right) = 0 \quad (\text{to find } x \text{ int})$$

$$\sin\left(\theta + \frac{\pi}{6}\right) = 0$$

$$\theta + \frac{\pi}{6} = 0, \pi, 2\pi$$

$$\theta = -\frac{\pi}{6}, \frac{5\pi}{6}, \frac{11\pi}{6}$$

$$\sqrt{3} \sin\left(\theta + \frac{\pi}{6}\right) = \sqrt{3}$$

$$\sin\left(\theta + \frac{\pi}{6}\right) = 1$$

$$\theta + \frac{\pi}{6} = \frac{\pi}{2}, \frac{5\pi}{2}$$

$$(\text{max pt}) \quad \theta = \frac{\pi}{3}$$

$$\sqrt{3} \sin\left(\theta + \frac{\pi}{6}\right) = -\sqrt{3}$$

$$\sin\left(\theta + \frac{\pi}{6}\right) = -1$$

$$\theta = \frac{3\pi}{2} - \frac{\pi}{6}$$

$$= \frac{4\pi}{3} \quad (\text{min pt})$$

MATHEMATICS Extension 1 : Question 12...

Suggested Solutions	Marks	Marker's Comments
a) $(1+x)^n = \binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + \dots + \binom{n}{n}x^n$ Differentiate with respect to x: $n(1+x)^{n-1} = \binom{n}{1} + \binom{n}{2}2x + \binom{n}{3}3x^2 + \dots + \binom{n}{n}nx^{n-1}$ Sub. $x=2$ $n(1+2)^{n-1} = \binom{n}{1} + \binom{n}{2}2(2) + \binom{n}{3}3(2)^2 + \dots + \binom{n}{n}n2^{n-1}$ $n \times 3^{n-1} = \binom{n}{1}(1) + \binom{n}{2}(2 \times 2) + \binom{n}{3}(3 \times 2^2) + \dots + \binom{n}{n}(n \times 2^{n-1})$ b) $\cot^2 \alpha + \cot^2 \beta + \cot^2 \gamma$ $= \operatorname{cosec}^2 \alpha - 1 + \operatorname{cosec}^2 \beta - 1 + \operatorname{cosec}^2 \gamma - 1$ $= \frac{1}{\sin^2 \alpha} + \frac{1}{\sin^2 \beta} + \frac{1}{\sin^2 \gamma} - 3$ $= \frac{\sin^2 \beta \sin^2 \gamma + \sin^2 \alpha \sin^2 \gamma + \sin^2 \alpha \sin^2 \beta}{\sin^2 \alpha \sin^2 \beta \sin^2 \gamma} - 3$ $(\sin \alpha \sin \beta + \sin \alpha \sin \gamma + \sin \gamma \sin \beta)^2 =$ $\sin^2 \alpha \sin^2 \beta + \sin^2 \alpha \sin \beta \sin \gamma + \sin \alpha \sin^2 \beta \sin \gamma$ $+ \sin^2 \alpha \sin \beta \sin \gamma + \sin^2 \alpha \sin^2 \gamma + \sin \alpha \sin \beta \sin^2 \gamma$ $+ \sin \alpha \sin^2 \beta \sin \gamma + \sin \alpha \sin \beta \sin^2 \gamma + \sin^2 \gamma \sin^2 \beta$ $= \sin^2 \alpha \sin^2 \beta + \sin^2 \alpha \sin^2 \gamma + \sin^2 \gamma \sin^2 \beta$ $+ 2 \sin^2 \alpha \sin \beta \sin \gamma + 2 \sin \alpha \sin^2 \beta \sin \gamma$ $+ 2 \sin \alpha \sin \beta \sin^2 \gamma$	1 1 1	

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$= \sin^2 \alpha \sin^2 \beta + \sin^2 \alpha \sin^2 \gamma + \sin^2 \gamma \sin^2 \beta$ $+ 2 \sin \alpha \sin \beta \sin \gamma (\sin \alpha + \sin \beta + \sin \gamma)$ $\therefore \sin^2 \alpha \sin^2 \beta + \sin^2 \alpha \sin^2 \gamma + \sin^2 \gamma \sin^2 \beta$ $= (\sin \alpha \sin \beta + \sin \alpha \sin \gamma + \sin \beta \sin \gamma)^2$ $- 2 \sin \alpha \sin \beta \sin \gamma (\sin \alpha + \sin \beta + \sin \gamma)$ $\therefore \cot^2 \alpha + \cot^2 \beta + \cot^2 \gamma =$ $\frac{(\sin \alpha \sin \beta + \sin \alpha \sin \gamma + \sin \beta \sin \gamma)^2 - 2 \sin \alpha \sin \beta \sin \gamma (\sin \alpha + \sin \beta + \sin \gamma) - 3}{(\sin \alpha \sin \beta \sin \gamma)^2}$ $= \frac{\left(-\frac{6}{8}\right)^2 - 2\left(\frac{\sqrt{5}}{8}\right)(0)}{\left(\frac{\sqrt{5}}{8}\right)^2} - 3$ $= \frac{21}{5}$	1 1	